

ACML100

Lecture 13

For a particle with Energy E in a region of potential energy V

$$\nabla^2 \psi(x, y, z) = -\left(\frac{2\pi}{\lambda}\right)^2 \psi(x, y, z)$$

$$\lambda = \frac{h}{[2m(E - V)]^{1/2}}$$

$$\left(\frac{2\pi}{\lambda}\right)^2 = \frac{4\pi^2 \times 2m \times (E - V)}{h^2} = \frac{8\pi^2 m(E - V)}{h^2}$$

$$E = \frac{1}{2}mv^2 + V = \frac{p^2}{2m} + V$$

$$p = [2m(E - V)]^{1/2}$$

$$\lambda = \frac{h}{p} = \frac{h}{[2m(E - V)]^{1/2}}$$

$$-\left(\frac{h^2}{8\pi^2m}\right)\nabla^2\psi(x,y,z)+V(x,y,z)\psi(x,y,z)=E\psi(x,y,z)$$

$$-\left(\frac{h^2}{8\pi^2m}\right)\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi(x,y,z)+V(x,y,z)\psi(x,y,z)=E\psi(x,y,z)$$

Postulates of Quantum Mechanics

Postulate 1:

State and wave functions. Born interpretation

The state of a quantum mechanical system is completely specified by a wave function $\psi(\mathbf{r}, t)$ that depends on the coordinates of the particles ($\mathbf{r}$) and time t . These functions are called wave functions or state functions.

For 2 particle system:

$$\Psi = \Psi(x_1, y_1, z_1, x_2, y_2, z_2, t)$$

Wave function contains all dynamic information about a system.

wave function	$\Leftrightarrow$	classical trajectory
(Quantum mechanics)		(Newtonian mechanics)

Postulates of Quantum Mechanics

Postulate 1:

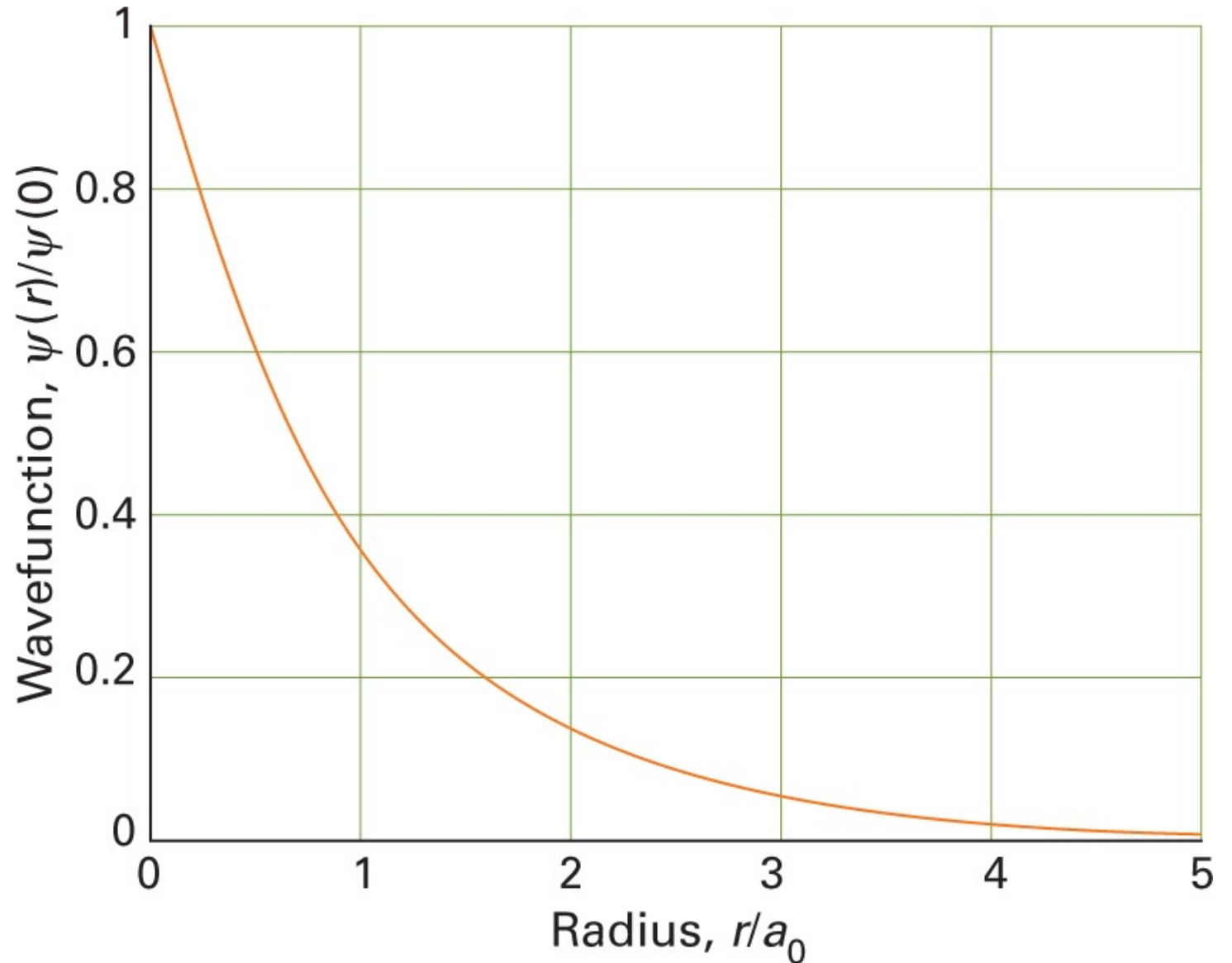
Meaning of wave function:

The probability of finding a particle in a small region of space of volume δV is proportional to $\psi^2 \delta V$ where ψ is the value of the wavefunction in the region.

$$P(r) \propto \psi^2 dV$$

=> the probability that the particle can be found at a particular point x and a particular time t . (Born's / Copenhagen interpretation)

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- The wavefunction of an electron in the lowest energy state of a hydrogen atom is proportional to e^{-r/a_0} , with $a_0 = 52.9$ pm

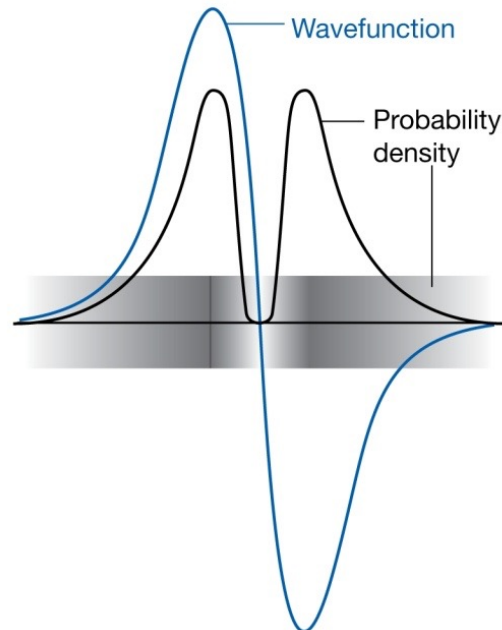


Implications of Born's Interpretation

(1) Positivity:

$$P(\mathbf{r}) \geq 0$$

The sign of a wavefunction has no direct physical significance:
The positive and negative regions of this wavefunction both correspond to the same probability distribution.



Normalization

The probability of finding the particle in the universe is 1.

$$\int_{all_space} \Psi^* \Psi d\tau = 1$$

$$\int_{-\infty}^{+\infty} \psi^* \psi d\tau = C \text{ (a constant)}$$

$$\int_{-\infty}^{+\infty} \frac{\psi}{\sqrt{C}} \frac{\psi^*}{\sqrt{C}} d\tau = 1$$

Normalization

$$\int_{-\infty}^{+\infty} \frac{\psi^*}{\sqrt{\infty}} \frac{\psi}{\sqrt{\infty}} d\tau = 1$$

$$\int_{-\infty}^{+\infty} \psi \psi^* d\tau = C < \infty$$

$$\int_{-\infty}^{+\infty} \psi \psi^* d\tau = \mathcal{C} < \infty$$

The wave function must be square-integratable. In other words, the integral of $|\Psi|^2$ over all space must be finite.

One consequence of this proposal is that Ψ must tend to 0 for infinite distances.

Physically acceptable wave function

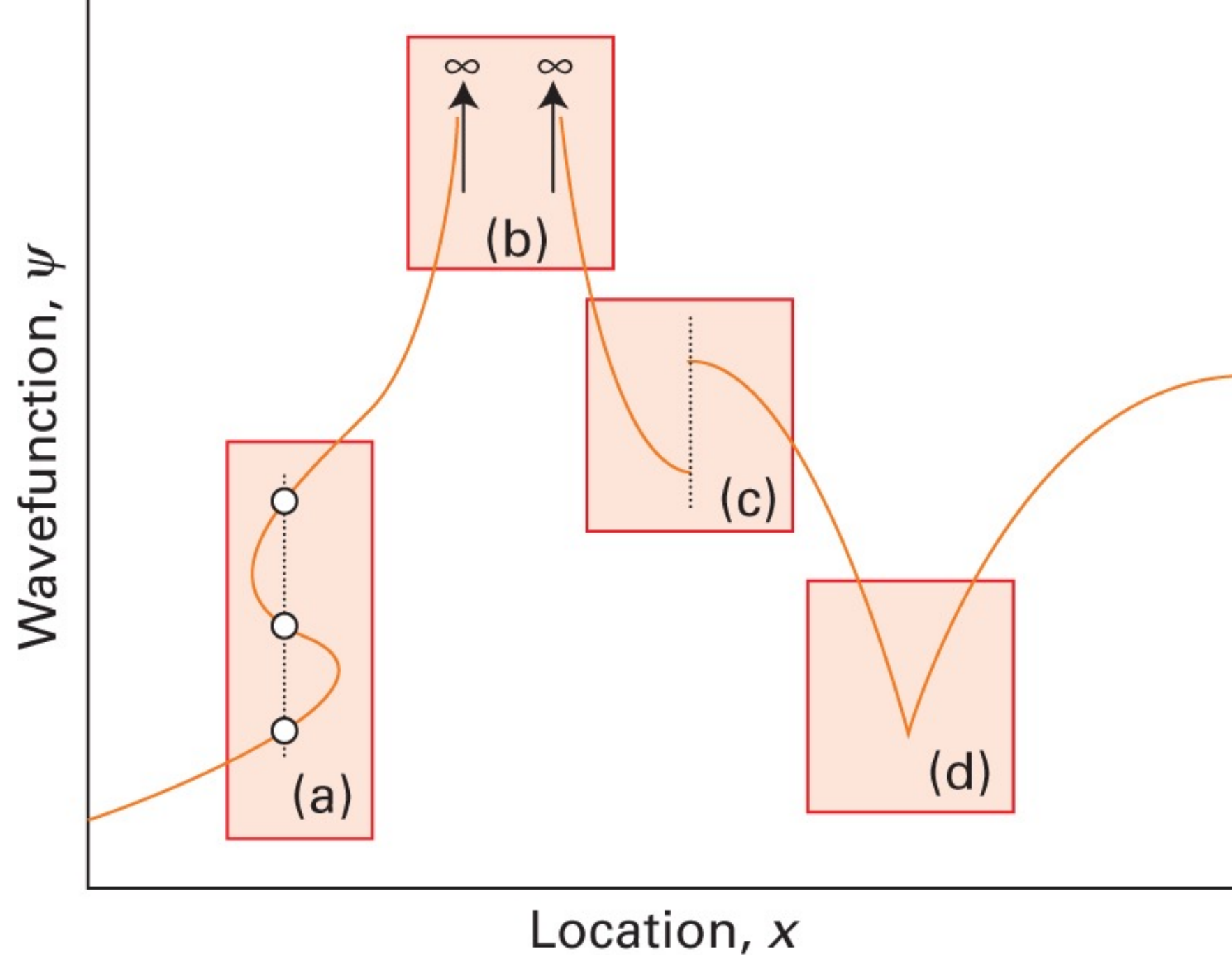
- The wave function and its first derivative must be:
 - 1) Finite. The wave function must be single valued. This means that for any given values of x and t , $\Psi(x,t)$ must have a unique value. This is a way of guaranteeing that there is only a single value for the probability of the system being in a given state.

Continuous wavefunction

- A rapid change would mean that the derivative of the function was very large (either a very large positive or negative number). In the limit of a step function, this would imply an infinite derivative. Since the momentum of the system is found using the [momentum operator](#), which is a first order derivative, this would imply an infinite momentum, which is not possible in a physically realistic system.

Continuous First derivative

1. All first-order derivatives of the wave function must be continuous. Following the same reasoning as in condition 3, a discontinuous first derivative would imply an infinite second derivative, and since the energy of the system is found using the second derivative, a discontinuous first derivative would imply an infinite energy, which again is not physically realistic.



Extracting information from the
wavefunction

Postulate 2

To every physical property, observable in classical mechanics, there corresponds a linear, hermitian operator in quantum mechanics.

Observables

- Anything which is function of position or/and momentum

Position: x

Momentum: p_x

$$\text{K.E.} : \frac{p_x^2}{2m}$$

$$\text{P.E.} : V(x)$$

$$\text{T.E} = \frac{p_x^2}{2m} + V(x)$$

Operator

- A rule that transforms a given function into another function

$$\hat{O}f(x) = g(x)$$

Operator

Operation	Operator	Function	Resulting Function
square	$(\)^2$	x^2	x^4
Square root	$\sqrt{\ }$	x^2	x
Multiplication by a constant	C	x^2	Cx^2
Differential (d/dx)	$\frac{d}{dx}$	x^2	$2x$
Second order differential	$\frac{d^2}{dx^2}$	x^2	2

Linear Operator

- A linear operator has the following properties

$$\hat{A}(f_1 + f_2) = \hat{A} f_1 + \hat{A} f_2$$

$$\hat{A}(cf) = c \hat{A} f$$

Operator $\hat{O}$	Function f_1	Function f_2	$\hat{O}(f_1 + f_2)$	$\hat{O}(f_1) + \hat{O}(f_2)$
$(\)^2$	x^2	x	$x^4 + x^2 + 2x^2$	$x^4 + x^2$
$\sqrt{\ }$	x^2	x	$\sqrt{x^2 + x}$	$x + \sqrt{x}$
$\frac{d}{dx}$	x^2	x	$2x + 1$	$2x + 1$
$\frac{d^2}{dx^2}$	x^2	x	2	2

Operator $\hat{O}$	Function f_1	constant	$\hat{O}(cf_1)$	$c\hat{O}(f_1)$
$(\)^2$	x^2	4	$16x^4$	$4x^4$
$\sqrt{\ }$	x^2	4	$2x$	$4x$
$\frac{d}{dx}$	x^2	4	$8x$	$8x$
$\frac{d^2}{dx^2}$	x^2	4	8	8

Hermitian operator

A hermitian operator $\hat{A}$ satisfies

$$\int f^* \hat{A} g dx = \int g (\hat{A} f)^* dx ; \text{ if } f \text{ and } g \text{ are well behaved}$$

Hermitian Operator

- Hermitian operators have two properties that forms the basis of quantum mechanics
 - (i) Eigen value of a Hermitian operator are real.
 - (ii) Eigenfunctions of Hermitian operators are orthogonal to each other or can be made orthogonal by taking linear combinations of them.

- Prove Operator x is Hermitian.

$$\int \psi_r^* \hat{x} \psi_s dx = \int \psi_s (\hat{x} \psi_r)^* dx$$

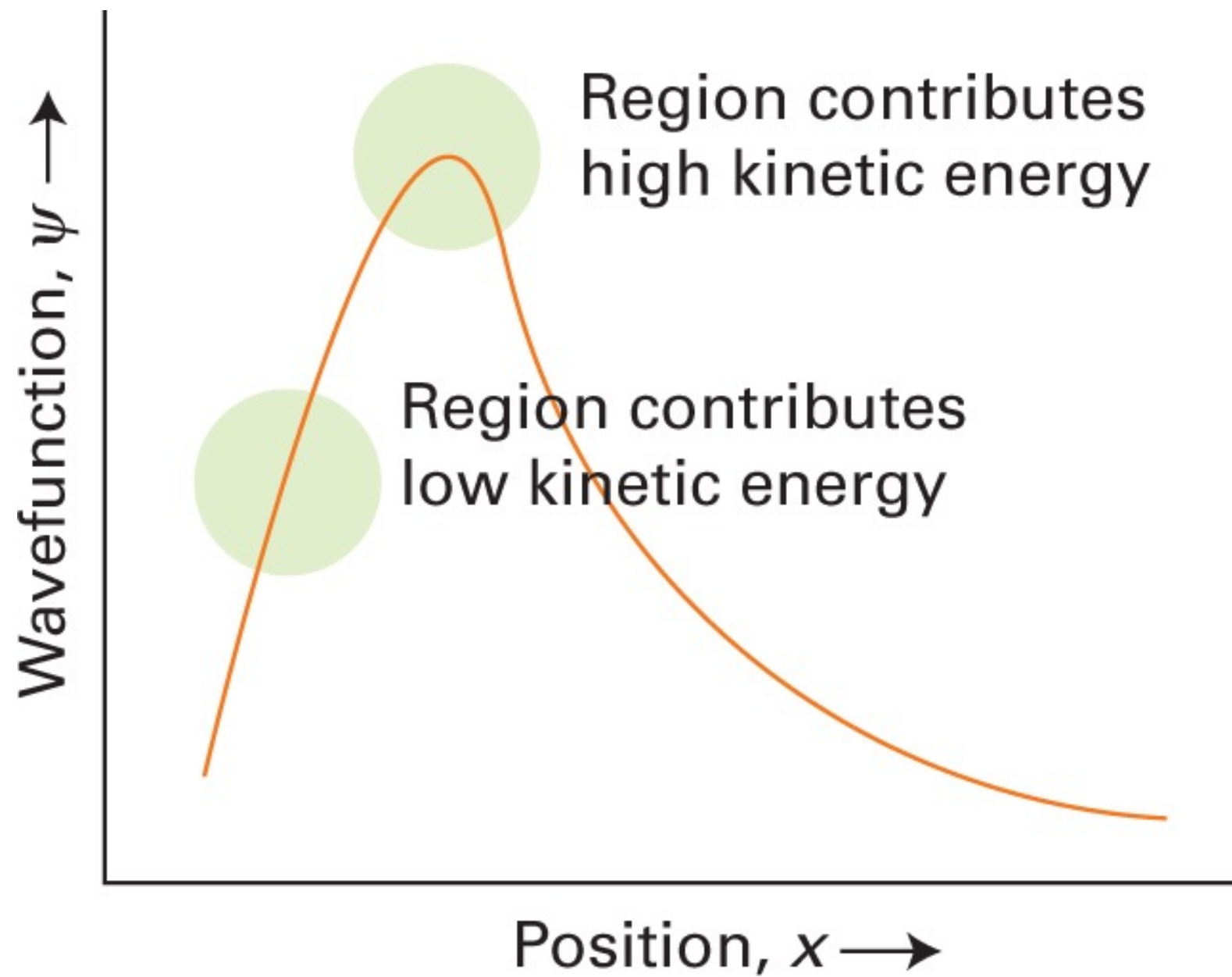
$$\int (\hat{x} \psi_r)^* \psi_s dx$$

$$= \int (x \psi_r)^* \psi_s dx$$

$$= \int \psi_r^* x \psi_s dx$$

$$= \int \psi_r^* \hat{x} \psi_s dx$$

Observable	Classical Symbol	Quantum Operator	Operation
Position	r	$\hat{r}$	Multiple by r
Momentum	p	$\hat{p}$	$-i\hbar \left(\hat{i} \frac{d}{dx} + \hat{j} \frac{d}{dy} + \hat{k} \frac{d}{dz} \right)$
Kinetic Energy	T	$\hat{T}$	$-\frac{\hbar^2}{2m} \left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right)$
Potential Energy	$V(r)$	$\hat{V}$	Multiply by $V(r)$
Total Energy	E	H	$-\frac{\hbar^2}{2m} \left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right) + V(r)$
Angular Momentum	l_x	$\hat{l}_x$	$-i\hbar \left(y \frac{d}{dz} - z \frac{d}{dy} \right)$



Postulate 3

In any measurement of the observable associated with the operator $\hat{A}$, the only values that will ever be observed are the eigenvalues 'a' which satisfy the eigenvalue equation:

$$\hat{A}\Psi = a\Psi$$

This is the postulate that the values of dynamical variables are quantized in quantum mechanics.

Eigen Function and Eigen value

$$\hat{A} f(x) = k f(x)$$

$f(x)$ is eigenfunction of $\hat{A}$ with eigen value k

Eigen function and eigen value

$$f(x) = e^{ikx}$$

Is it eigen function of momentum operator ?

What is eigen value ?

Eigenvalue equation

Eigenvalue equation

(Operator)(function) = (constant factor)*(same function)

Example: e^{ikx} is an eigenfunction of a operator $\hat{P}_x = -i\hbar \frac{\partial}{\partial x}$

$$f(x) = e^{ikx}$$

$$= -i\hbar \frac{\partial}{\partial x} e^{ikx}$$

$$= -i^2 \hbar k^2 e^{ikx}$$

$$= \hbar k^2 e^{ikx} \quad \text{Thus } e^{ikx} \text{ is an eigenfunction}$$

Predicting the outcome of
experiments

Postulate 4

For a system in a state described by a normalized wave function Ψ , the average or expectation value of the observable corresponding to A is given by:

$$\langle A \rangle = \int \Psi^* \hat{A} \Psi d\tau$$

Mean value theorem

Expectation value in general:

$$\begin{aligned}\langle a \rangle &= \frac{\int_{-\infty}^{\infty} \Psi^*(x) \hat{A} \Psi(x) dx}{\int_{-\infty}^{\infty} \Psi^*(x) \Psi(x) dx} \\ &= \int_{-\infty}^{\infty} \Psi^*(x) \hat{A} \Psi(x) dx \quad (\text{if normalized})\end{aligned}$$

The fourth postulates states what will be measured when large number of identical systems are interrogated one time. Only after large number of measurements will it converge to $\langle a \rangle$.